Class 10 Science - Chapter 9

Light - Reflection and Refraction (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)
Class 7-8 ki yeh cheezein pehle pakki karo, warna Ch9 samajh nahi aayega.

1. Light ka basic (Class 7-8)

Light = energy ka ek roop jo seedhi line me chalti (rectilinear propagation).
Reflection (paraavartan) = light ka kisi surface se TAKRAKAR wapas aana.
Mirror = chamakdar surface jo light reflect kare.
Image (pratibimb) = kisi cheez ka mirror/lens se bana roop.

2. Angle aur Normal (Class 7 geometry)

Normal = surface par 90 degree (perpendicular) wali kalpit (imaginary) line.
Incident ray = aane wali light. Reflected ray = wapas jaane wali light.
Angle of incidence (i) = incident ray aur normal ke beech ka kona.
Angle of reflection (r) = reflected ray aur normal ke beech ka kona.

3. Real vs Virtual image

Real image = screen par pakdi ja sakti (jaha kiranein actually milti).
Hamesha ULTI (inverted) hoti.
Virtual image = screen par nahi pakdi ja sakti (kiranein sirf aage badhti lagti). Hamesha SEEDHI (erect) hoti. (Jaise plane mirror me apna chehra.)

4. Concave vs Convex shape

Concave = andar dhasaa hua (jaise chammach ka andar wala part).
Convex = bahar ubhaar (chammach ka peeche/baahar wala part).
Transparent = aar-paar dikhe (kaanch); medium = jis me se light guzre.

5. Maths basics (numericals ke liye)

Sign ka dhyan: +ve aur -ve number. Formula me 1/f, 1/v, 1/u aate.
Unit: distance cm/m me. Reciprocal (1/x) lena aata ho.
Ratio/cross-multiply se v ya u nikalna.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT
Class 10 Science | Ch 9: Light - Reflection and Refraction

[GREEN = board exam me baar-baar poocha jaata hai]

1. Laws of Reflection [IMPORTANT]

(1) Angle of incidence = Angle of reflection (i = r).
(2) Incident ray, reflected ray aur normal - teeno ek hi plane me hote.
Plane mirror ka image: virtual, erect (seedha), same size, mirror ke peeche utni hi door, aur LATERALLY INVERTED (left-right ulta - jaise AMBULANCE ka likha ulta).

2. Spherical Mirrors [TOP EXAM AREA]

CONCAVE mirror = reflecting surface ANDAR (converging - kiranein ek point par milati). CONVEX mirror = reflecting surface BAHAR (diverging - kiranein failati).
Important terms:
Pole (P): mirror ka center point.
Centre of curvature (C): jis sphere ka mirror hissa, uska center.
Radius of curvature (R): P se C ki doori.
Focus (F): kiranein (axis ke parallel) reflect hokar jahan milti/lagti.
RELATION: R = 2f (focus = radius ka aadha).

3. Image by Concave Mirror (object ki position se) [VERY IMPORTANT]

Object kahan	Image kahan	Nature + Size
Infinity	F par	Real, inverted, point
Beyond C	C aur F ke beech	Real, inverted, chhota
At C	At C	Real, inverted, same size
C aur F ke beech	Beyond C	Real, inverted, bada
At F	Infinity	Real, inverted, bahut bada
F aur P ke beech	Mirror ke peeche	Virtual, erect, bada

Concave mirror uses: shaving/makeup (paas se bada seedha), torch/headlight reflector, dish antenna, doctor ka head mirror, solar furnace.

4. Image by Convex Mirror

Convex mirror se image HAMESHA: virtual, erect, CHHOTA (diminished), mirror ke peeche (F aur P ke beech). Object kahin bhi ho.
Uses: gaadi ka SIDE/REAR-view mirror (chhota image -> bada area dikhta, wider field of view), dukaan/blind turns par security mirror.

5. Mirror Formula aur Magnification [TOP EXAM - NUMERICALS]

Mirror formula: $1/v + 1/u = 1/f$
v = image distance, u = object distance, f = focal length.
Magnification: $m = h'/h = -v/u$
h' = image height, h = object height.
Sign convention (New Cartesian):
- Saari doori POLE se naapo.
- Mirror ke saamne (left) = NEGATIVE; peeche (right) = POSITIVE.
- Object hamesha saamne -> u hamesha NEGATIVE.
- Upar height +ve, neeche -ve.
- Concave mirror ka f = NEGATIVE; Convex ka f = POSITIVE.

m ka matlab: m -ve -> real+inverted; m +ve -> virtual+erect.
$|m|>1$ bada, $|m|<1$ chhota, $|m|=1$ same size.

6. Refraction of Light [TOP EXAM AREA]

Refraction = light ka ek medium se doosre me jaane par MUD-na (bend hona),
kyunki speed badal jaati.
Rules:
- Kam dense (rare) se zyada dense me -> normal ki TARAF mudti.
- Zyada dense se kam dense me -> normal se DOOR mudti.
- Normal par seedha (90 par) aaye -> nahi mudti.

Glass slab se light parallel shift hoti (aati hui aur jaati hui ray parallel).
Common observation: paani me dali pencil tedhi dikhti; pool kam gehra dikhta;
tare timtimate (atmospheric refraction).

7. Refractive Index [IMPORTANT]

Refractive index (n) = light medium me kitna dheere chalti, uska maap.
n = (speed of light in vacuum c) / (speed in medium v) = c/v.
Zyada n = zyada dense (optically) = light zyada dheere + zyada mudti.
Eg: water n=1.33, glass n=1.5, diamond n=2.42 (sabse zyada -> bahut
chamakta, light andar phasati).

8. Lenses (Refraction se) [TOP EXAM AREA]

CONVEX lens = beech me MOTA, kinare patle. CONVERGING (kiranein milati).
CONCAVE lens = beech me PATLA, kinare mote. DIVERGING (kiranein failati).
Optical centre (O), Principal focus (F), focal length (f), 2F.

Convex lens image (object position se):
Beyond 2F -> 2F aur F ke beech, real inverted chhota.
At 2F -> at 2F, real inverted same size.
2F aur F ke beech -> beyond 2F, real inverted bada.
At F -> infinity. Between F aur O -> same side, virtual erect bada (magnify).
Concave lens: hamesha virtual, erect, chhota (object aur lens ke beech).

Lens formula: $1/v - 1/u = 1/f$
Magnification (lens): $m = h'/h = v/u$
Sign: convex lens f = +ve; concave lens f = -ve. u object ka -ve.

9. Power of Lens [IMPORTANT]

Power P = 1/f (f METRE me). Unit = DIOPTRE (D).
Convex lens power = +ve; Concave lens power = -ve.
f chhoti -> power zyada (zyada mudaata).
Lens combination: total power P = P1 + P2 + ... (powers jod do).

10. Quick Revision - One Liners

- i = r (reflection). Plane mirror: virtual, erect, laterally inverted.
- R = 2f. Mirror: 1/v+1/u=1/f, m=-v/u. Lens: 1/v-1/u=1/f, m=v/u.
- Concave mirror f -ve, convex +ve; convex lens f +ve, concave -ve.
- Convex mirror + concave lens: hamesha virtual, erect, chhota.
- Power = 1/f(m), unit dioptre; n = c/v.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Concave mirror numerical (full sign convention)

Q: Ek concave mirror ki focal length 15 cm. Object 10 cm door rakha. Image distance (v) aur magnification (m) nikaalo. Nature batao.

Solution:
- f = -15 cm (concave), u = -10 cm (saamne).
- Mirror formula: 1/v + 1/u = 1/f -> 1/v = 1/f - 1/u = 1/(-15) - 1/(-10).
- 1/v = -1/15 + 1/10 = (-2 + 3)/30 = 1/30 -> v = +30 cm.
- v +ve -> image mirror ke PEECHE -> virtual, erect.
- m = -v/u = -(30)/(-10) = +3 -> erect, 3 guna BADA.

(Object F aur P ke beech tha, isliye virtual+erect+bada - table se match.)

Example 2 - Convex lens numerical

Q: Convex lens f = 10 cm, object 15 cm door. v aur m nikaalo.

Solution:
- f = +10 cm (convex), u = -15 cm.
- Lens formula: 1/v - 1/u = 1/f -> 1/v = 1/f + 1/u = 1/10 + 1/(-15).
- 1/v = 1/10 - 1/15 = (3 - 2)/30 = 1/30 -> v = +30 cm.
- m = v/u = 30/(-15) = -2 -> real, inverted, 2 guna bada.

Example 3 - Power of lens combination

Q: +5 D aur -2 D ke do lens jode. Net power aur focal length?

Solution:
- Net power P = P1 + P2 = +5 + (-2) = +3 D.
- f = 1/P = 1/3 m = 0.33 m = 33.3 cm (convex, kyunki +ve).

Example 4 - Refractive index

Q: Glass ka refractive index 1.5. Vacuum me light speed 3x10^8 m/s. Glass me speed kya?

Solution:
- n = c/v -> v = c/n = (3x10^8)/1.5 = 2x10^8 m/s.

Example 5 (EASIEST) - Mirror choose karna

Q: Gaadi ka rear-view (side) mirror kaunsa hota aur kyun?

Solution:
- CONVEX mirror. Kyunki yeh chhota+seedha image deta aur bada area (wider

field of view) dikhata -> peeche ka traffic zyada dikhe.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

Q1. Reflection ke 2 niyam.
Q2. R aur f ka relation kya hai?
Q3. Convex mirror ka image hamesha kaisa hota?
Q4. Power of lens ka formula + unit.

MEDIUM (2-3 marks)

Q5. Real aur virtual image me 3 antar.
Q6. Concave mirror ke 3 uses + kyun.
Q7. Refraction ke rules (dense/rare medium me kaise mudti).
Q8. Mirror formula aur magnification formula + sign convention.

HARD (3-5 marks, board favourite)

Q9. Concave mirror f=15cm, object u=-10cm -> v aur m + nature.
Q10. Convex lens f=10cm, object 15cm -> v aur m + nature.
Q11. Concave mirror se object ki har position ka image (table).
Q12. Refractive index kya? n=c/v se glass me speed (n=1.5) nikaalo.

ANSWER HINTS (PART D)

A1. i=r ; incident, reflected, normal ek plane me.
A2. R = 2f.
A3. Virtual, erect, chhota (diminished).
A4. P = 1/f(metre); unit dioptre (D).
A5. Real: screen par, inverted; Virtual: screen par nahi, erect.
A6. Shaving/makeup (bada seedha), torch reflector, solar furnace.
A7. Rare->dense: normal ki taraf; dense->rare: normal se door.
A8. 1/v+1/u=1/f, m=-v/u; saamne -ve, peeche +ve, concave f -ve.
A9. v=+30cm, m=+3 -> virtual, erect, 3x bada.
A10. v=+30cm, m=-2 -> real, inverted, 2x bada.
A11. Infinity->F; beyond C->C-F chhota; at C->same; C-F->beyond C bada;
F->infinity; F-P->virtual erect bada.
A12. n=c/v; v=c/n=3x10^8/1.5=2x10^8 m/s.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!
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